

LECTURE ACTIVITY 1: DON'T GET RENT OUT OF SHAPE

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Learning Objectives:

By the end of this activity, you should be able to

- 1) Define and identify observational units.
- 2) Define and identify variables.
- 3) Classify variables as categorical or quantitative.
- 4) Extract basic information from simple data visualizations.
- 5) Manipulate variables within plots.

Key Terms:

- | | | |
|----------------------|-----------------------------|--------------------|
| • Data Table | • Variable (Categorical vs. | • Barplot/Bargraph |
| • Data Dictionary | Quantitative) | • Histogram |
| • Observational Unit | • Exploratory Data Analysis | • Scatterplot |
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Welcome to our first STAT 1110 Lecture Activity! Lecture Activities are designed to give you a more hands-on approach to learning the material in this course. Feel free to work in groups; there is a lot we can learn from one another! Also, no need to turn this in (though I highly encourage you to complete it!)

Here is a quick re-introduction to some key statistical terms and concepts we learned earlier today. We think of **data** as a collection of values recorded on individuals or objects of interest (called **observational units**). These values are often measurements of various attributes/characteristics, which we call **variables**. At the highest level, variables fall into one of two types: **categorical** or **quantitative**. Categorical variables are those whose values are one of a fixed set of categories (called **levels**), and quantitative variables are those whose values are numbers.

More accurately; quantitative variables are those for which it makes “interpretive sense” to perform arithmetic operations. For example, even if we record “month” using 01 for January, 02 for February, etc., the variable “month” does not become categorical - that’s because “01” + “02” (i.e. “January” + “February”) has no interpretive meaning!

So, when determining whether a variable is categorical or quantitative, ask yourself: “does it make interpretive sense to perform arithmetic operations on this variable?”

Introduction to the Dataset

*Zillow*¹ is a popular online database that lists houses and apartments for sale or rent. I (Dr. Ethan) collected the following data from Zillow in late June of this year (2026), by searching for a handful of apartment rentals in San Luis Obispo, CA. Each listing corresponds to a unique apartment, and only apartments were included (i.e. houses and townhomes were excluded from the dataset.) The variables included in this particular dataset are:

¹<https://www.zillow.com/>

- **Price:** The rental price (in USD per month)
- **NumBed:** The number of bedrooms (Studio apartments are listed as having “0” bedrooms)
- **NumBath:** The number of baths (including half-baths)
- **Pool:** Whether or not the property has a pool (1 = has a pool, 0 = does not have a pool)
- **Size:** The size (in square feet)

Recall that a specification of all variables along with a short description of each (like what we have above) is what we refer to as a **Data Dictionary**.



Question 1

Based on the data dictionary, classify each of the variables as either quantitative or categorical.

- | | | |
|-------------------|-----------------------------------|------------------------------------|
| • Price: | ☐ Categorical | ☐ Quantitative |
| • NumBed: | ☐ Categorical | ☐ Quantitative |
| • NumBath: | ☐ Categorical | ☐ Quantitative |
| • Pool: | ☐ Categorical | ☐ Quantitative |
| • Size: | ☐ Categorical | ☐ Quantitative |

Here are the first few rows of the **Data Table**:

Price	NumBed	NumBath	Pool	Size
599	1	1	1	185
599	1	1	1	93
799	1	1	1	122
899	1	1	1	122
899	1	1	1	185
949	1	1	1	185
<i>The table continues, and contains 82 rows in total</i>				

Table 1: First Few Rows of the Data Table



Question 2

What are the observational units in the dataset? How many observational units are there? How can you tell?

Exploratory Data Analysis

Many statistics project begins with what is known as **exploratory data analysis (EDA)**. As the name suggests, EDA is dedicated to exploring the dataset before diving into more complex analyses. Indeed, a big part of EDA is generating informative **visualizations** (i.e. **plots**).

Recall that the two types of variables (categorical and quantitative) lend themselves more naturally to different types of plots.

- Categorical variables lend themselves to **barplots/bargraphs**, in which we have one bar for each level and the height of each bar is proportional to the frequency of the corresponding level
- Quantitative variables lend themselves to **histograms**, in which we see how many observations fall into a set of **bins**, each of a fixed (and predetermined) **binwidth**.
 - Quantitative variables with a small number of observations also lend themselves to **dotplots**. We'll talk about the differences between histograms and dotplots a little later.

Well-crafted visualizations can tell us a lot about data! Let's get some practice extracting information from graphs. (We'll talk more about *creating* graphs next lecture.) First, let's get a sense of the **distribution** of properties that have/don't have pools. Based on the classification of the "Pool" variable, we know to make a bargraph:

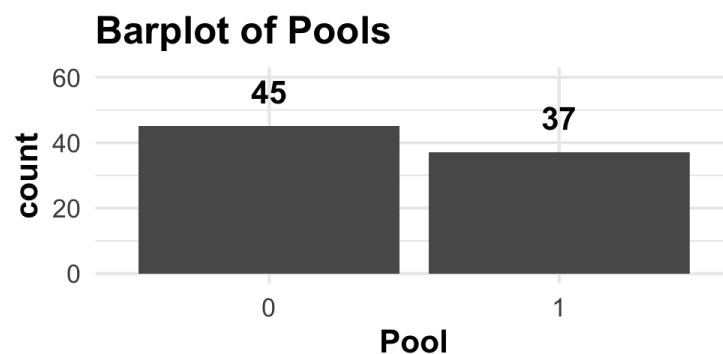


Figure 1: Bargraph of the distribution of which apartments in the dataset have pools and which do not.

Question 3

Which are more common in the dataset: apartment complexes with pools, or apartment complexes without pools? By how much do the counts of these complexes differ - is one type of complex drastically more represented than the other? Explain briefly.

Now, let's turn our attention to the prices of apartments. Since "Price" is a quantitative variable, we can visualize its distribution using a histogram:

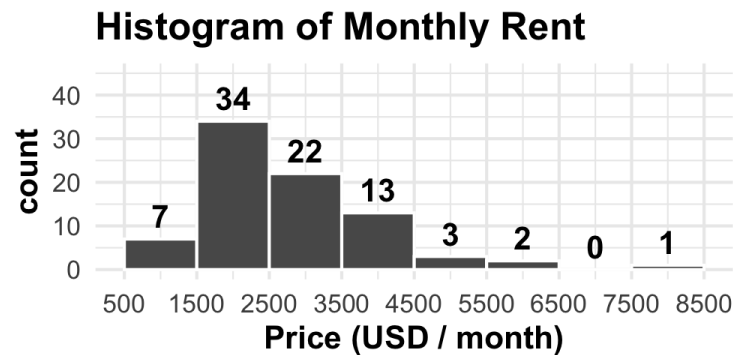


Figure 2: Histogram of the distribution of apartment prices (monthly rents).



Question 4

- (a) How many apartments in the dataset have rents between \$500 and \$1500 per month? Or, can this information not be determined from the histogram in Figure (2)?

- (b) How many apartments in the dataset have rents between \$1500 and \$2500 per month? Or, can this information not be determined from the histogram in Figure (2)?

- (c) How many apartments in the dataset have rents exactly equal to \$1200 per month? Or, can this information not be determined from the histogram in Figure (2)?

- (d) What is the most common price range, as determined from the histogram in Figure (2)? What about the least common?

Next, let's explore the distribution of the number of bedrooms. Because **NumBed** is quantitative, we use a histogram:

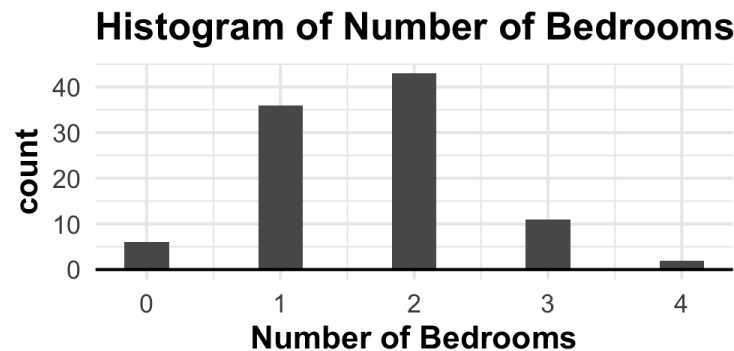


Figure 3: Histogram of the distribution of the number of bedrooms.

 Question 5

Why does this histogram have “gaps?” More specifically, what about the **NumBed** variable is qualitatively different from, say, the **Price** variable we explored in the previous question? **Hint:** think about the possible values the **NumBed** variable could take, and compare that with the possible values the **Price** variable could take.

We'll expand upon this point a bit more next lecture (and talk about how a different type of plot, called a **dotplot** could have been utilized here). For now, even though the histogram in Figure (3) looks a little different, we can still interpret it like any histogram.

 Question 6

- (a) What is the most frequently occurring number of bedrooms? Approximately how many apartments in the dataset have this number of bedrooms?

- (b) What is the least frequently occurring number of bedrooms? Approximately how many apartments in the dataset have this number of bedrooms?

More Involved Plots

Now that we have a feel for extracting information from histograms and bargraphs, let's take this a bit further! Specifically, we'll work toward answering the following questions through a single plot:

- What, if any, is the relationship between the size of an SLO-area apartment and its rental price?
- Does whether or not the apartment have a pool affect this relationship?
- How, if at all, does the number of bedrooms in the apartment affect this relationship?

First, let's focus on answering the first question: "what, if any, is the relationship between the size of an SLO-area apartment and its rental price?" This is a question about the relationship between the **Size** and **Price** variables, which is a comparison between two quantitative variables. As such, a **scatterplot** is best suited to answer this question.

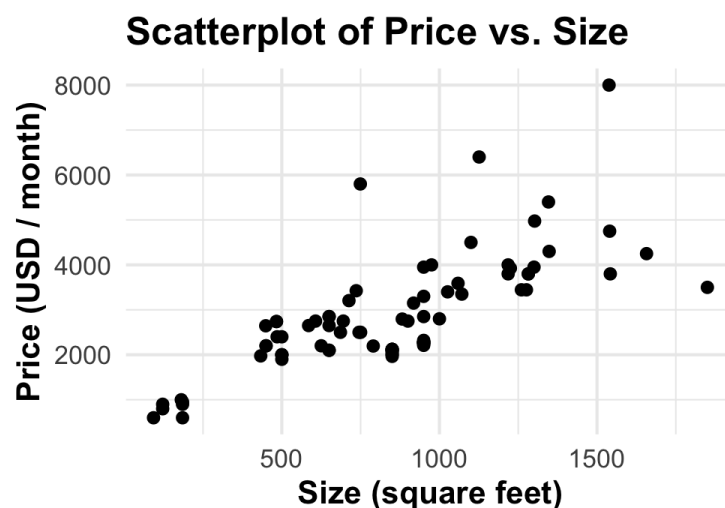


Figure 4: Scatterplot comparing the size and prices of apartments included in the dataset.

Question 7

What (if any) is the relationship between an apartment's size and its rent? Are larger apartments more or less expensive than smaller apartments? Is the relationship (if it exists) linear, or nonlinear - that is, do the points seem to fall along a line, or do they more closely fall along a curve?

Hm, I wonder if the relationship between an apartment's price and its square footage changes depending on whether or not the apartment has a pool? Note that each point in Figure 4 above represents an indi-

vidual apartment. As such, we can **change the shape of each point depending on whether or not the associated complex has a pool!**

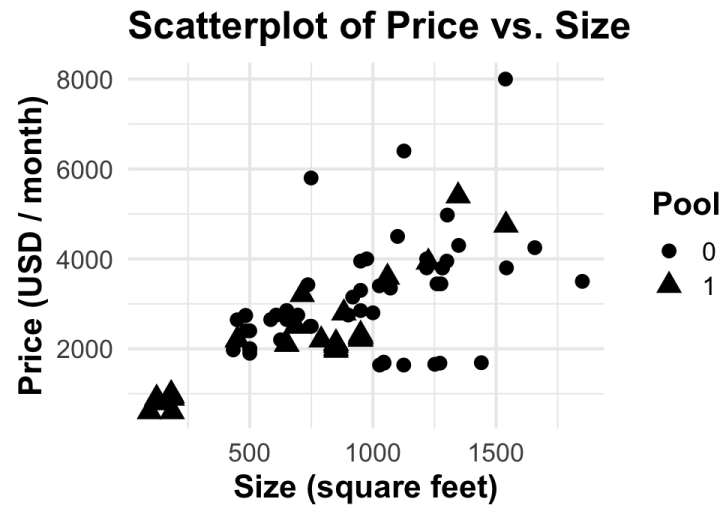


Figure 5: Scatterplot comparing the size and prices of apartments included in the dataset, with each point's shape corresponding to whether or not the apartment has a pool.



Question 8

Does the relationship you described in Question (7) above change depending on whether or not the associated apartment complex has a pool? Furthermore, are apartments with pools significantly more expensive than those without? Are they significantly bigger? How can you tell?

Finally, we ask the same question about the number of bedrooms. In addition to changing the shape of each point depending on whether or not the associated complex has a pool, we can **change the color of each point to reflect the number of bedrooms.**

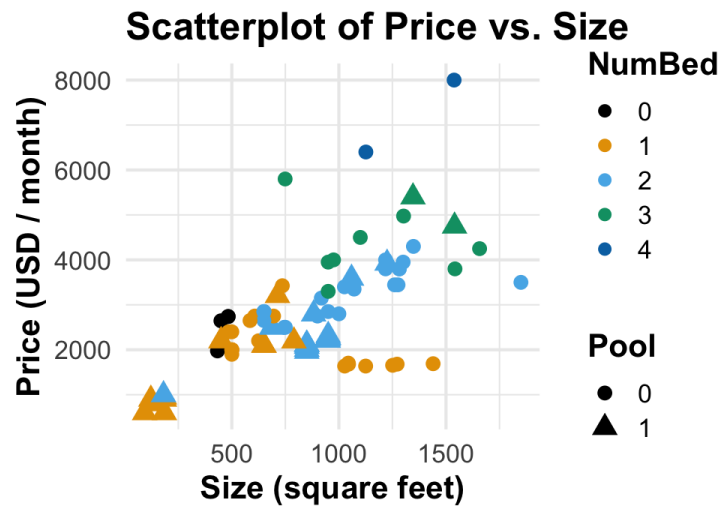


Figure 6: Scatterplot comparing the size and prices of apartments included in the dataset, with each point's shape corresponding to whether or not the apartment has a pool and each point's color corresponding to the number of bedrooms.



Question 9

Describe the relationship between the four variables pictured in Figure (6). Consider questions like: “do bigger apartments tend to have more bedrooms?” and “among apartments with pools, are bigger apartments more expensive than smaller ones?”

On Wednesday, we'll talk more about plots and good practices when it comes to generating them. One thing I'll leave you off with now is: take another look at Figure (6). It contains a lot of information - in fact, I might even say it borders on containing too much information. This is something I'll be stressing a lot throughout this course: oftentimes, simplicity is key. It's important to take care not to overwhelm the reader with any one plot. Again, more about this on Wednesday!

For now, I'd like to close out with one more question.

 Question 10

Consider the following histogram of the **NumBath** variable (which, you may recall, tracks the number of bathrooms in each apartment).

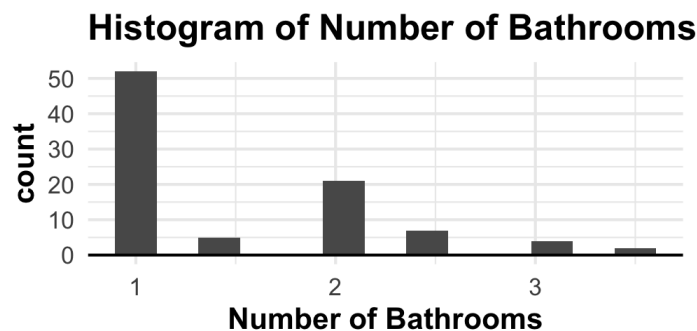


Figure 7: Histogram of the number of bathrooms. Note the presence of half-baths.

A (fictional) student has written the following interpretation.

The histogram reveals that the most common number of bathrooms is 1, followed by 2, and 3. The presence of half-baths is somewhat rare. This is a phenomenon we can conclude must hold in apartments across the United States.

Do you agree with the student's reasoning? In particular, if you disagree, what part specifically do you disagree with?